

2026 UNHCR Data and Innovation for Refugee Inclusion Hackathon

*The Oversight Layer: A Judging Engine and
Cognitive-Forcing Interface for Appropriate
Human Reliance on AI*

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2 October 2026

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1 Context

UNHCR gives cash to forcibly displaced households so they can cover basic needs such as food, rent and medicine. Funding never covers every household that needs help, so each operation has to target: decide who receives cash, and in what order. In the UNHCR operation in Latin America behind this challenge, targeting follows five steps:

1. **Interview:** Registration staff interview the household (who they are, documents, health housing, basic needs and how they cope). The interviewer also records their own view of the case;
2. **Scorecard:** An expert-designed Scorecard turns the answers into a final score and a vulnerability category, and runs administrative checks (asylum procedure, stated intentions, duplicate registration);
3. **AI recommendation:** Cashy, a UNHCR prototype, predicts the score and category and recommends include or exclude, with a written reasoning beside it;
4. **Decision:** A caseworker reviews the case and accepts or overrides the recommendation. The household is not present and cannot contest it;
5. **Prioritized cash distribution** Eligible households receive cash in order of priority, as far as funds allow. Others go to a waiting list or are excluded.

2 Challenge Objective

Solutions may address either question. The second is the core of the challenge: a better model helps, but it will still be wrong sometimes, and a caseworker has to catch it when it is:

Question 1. Can we predict better which households should receive cash?

Question 2. How do we make sure caseworkers keep overriding the AI when its advice is wrong, and that the institution can see when they stop?

We have focused on question 2 to find a solution.

3 Our solution: The Oversight Layer

Model errors are unavoidable, but the real risk is an operator that blindly accepts a wrong recommendation. The focus is on how humans interact with the system, rather than how the model works.

Operator trust is inherently unstable. Users exhibit *algorithm appreciation* by preferring AI predictions to human judgment [1], yet quickly switch to *algorithm aversion* after observing a single error [2]. Simply increasing transparency does not solve this: displaying AI reasoning inflates trust regardless of accuracy [3], leading to automation bias where operators passively accept incorrect advice [4].

Meaningful user engagement requires an active process rather than a single click [5]. Instead of retraining Cashy, we added an *Oversight Layer* between model outputs and human decisions, to prioritize the caseworker’s critical thinking over automation.

Grounded in cognitive forcing mechanisms [6], the system will:

- Treat Cashy’s reasoning and final recommendation as separate claims;

- Flag record discrepancies
- Mandate a brief survey on every case
- Track reliance patterns across the institution.

3.1 Dataset Analysis

The first step of our work was the *dataset analysis*.

During the analysis performed on the `S8.synthetic_cashy_sample.csv`, we observed that eligibility determination is not well predicted by the Scorecard factors alone; the Cashy model takes other inputs that are not included in the dataset.

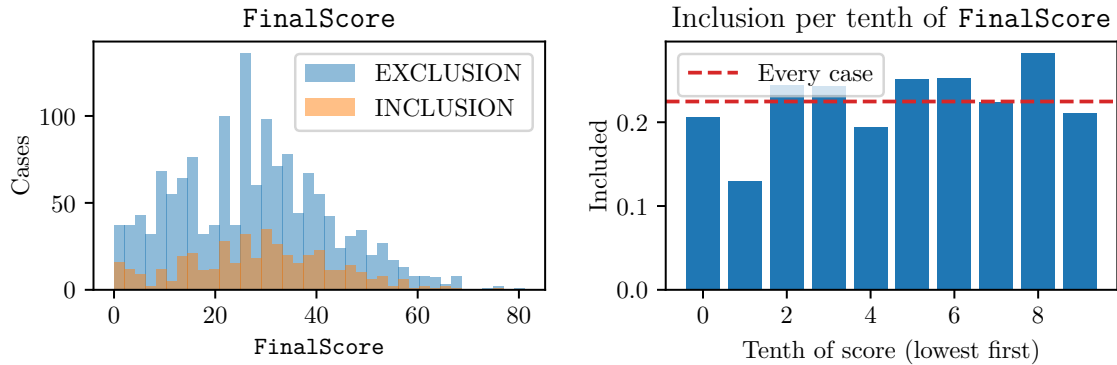


Figure 1: **FinalScore** distribution and inclusion rate across deciles.

As you can see in Figure 1, the **FinalScore** distribution for the included cases strongly overlaps with that of the excluded cases. Dividing the dataset into score deciles shows that the inclusion rate fluctuates around the overall average of 22.5% regardless of the score bracket, producing an *Area Under the Curve (AUC)* of only 0.529. This confirms that score factors alone do not determine the final eligibility outcomes.

3.2 Workflow Pipeline

The workflow pipeline is still pretty much the same; we only added a new process between the *AI recommendation* and *decision* processes and slightly modified the other ones.

Our solution works as follows (see the diagram in Figure 2):

- **Interview:** Registration staff interview the household: who they are, documents, health, housing, basic needs and how they cope. The interviewer also records their own view of the case. Both scorecard and Cashy will be fed those information;
- **Cashy:** Cashy, a UNHCR AI prototype, predicts the score and category and recommends **Include** or **Exclude**, with a written reasoning beside it;
- **Judgement process:** A series of functions written in Python use an AI model to compare the scorecard with the score given by Cashy to return a % value. This value is compared with a threshold value of 75 (check `judgement.ipynb` for more details on the implementations and the reasoning behind the threshold value);
- **Operator Survey:** Improvement of the already existing post-session questionnaire. The new questions are:

- Rate the effectiveness of Cashy-AI for identifying and selecting the “right” individuals for the assistance;
- Do you agree or disagree with Cashy-AI’s judgement? Why?
- Did Cashy-AI provide content that was relevant to help you make an informed decision?

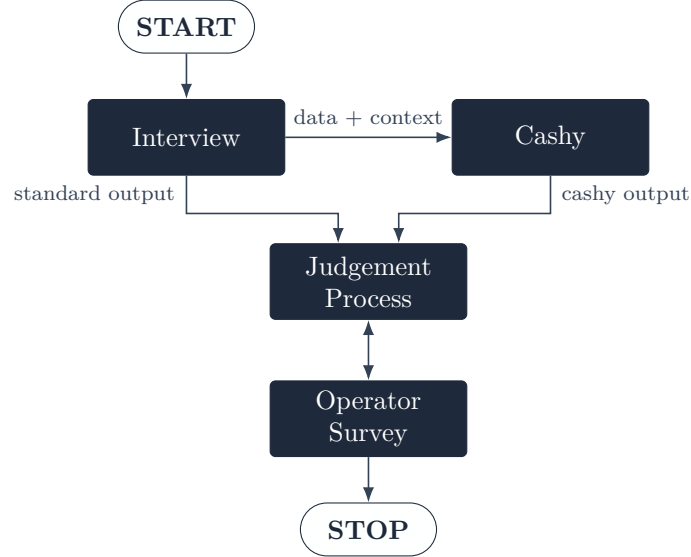


Figure 2: Workflow pipeline diagram.

3.3 Dashboard

The core objective is to provide a transparent and bias-aware interface where human case-workers can review AI-generated eligibility suggestions, understand the underlying reasoning [3], and make independent decisions free from automation bias [4].

Our solution’s primary strength is its *bias awareness*: the warning logic and critical case modal demonstrate a proactive approach to identifying potential AI bias, employing cognitive forcing mechanisms to trigger the operator’s critical thinking [6] and sustain active user engagement [5].

3.3.1 Technology Stack

The backend is built with Python and Flask, providing a REST API for retrieving case data and submitting operator decisions. The API uses JSON-formatted request and response bodies. The frontend is built with HTML, JavaScript, and CSS, with Tailwind for styling.

Following the API endpoints:

Table 1: API endpoints

Category	Method	Endpoint	Description
Data Retrieval	GET	/api/case/pending	List of unreviewed cases
Data Retrieval	GET	/api/cases	Case summaries for the sidebar selector

Continued on next page

Category	Method	Endpoint	Description
Data Retrieval	GET	/api/cases/<id>	Context checklist and metadata
Data Retrieval	GET	/api/case/<id>	Raw scorecard JSON (not displayed in UI)
Data Retrieval	GET	/api/cashy/<id>	Raw Cashy AI JSON (not displayed in UI)
Data Retrieval	GET	/api/comparison/<id>	Core judgement result (differences, semantic match, bias risk)
Operator Actions	POST	/api/case/<id>/review	Submit an independent operator decision
Operator Actions	PUT	/api/case/<id>/review	Revise a decision before survey completion
Survey & Feedback	GET	/api/survey/<id>	Survey status (currently disabled)
Survey & Feedback	POST	/api/survey/<id>	Submit survey (currently returns 501)

3.3.2 User Interface Layout

As shown in Figure 3, the *user interface* follows a three-pane design:

- **Left Sidebar:** Case list with status indicators (Pending, Awaiting survey). Allows quick navigation between cases.
- **Center Pane:** *External variables* accordion and *Cashy suggestion* summary. This provides the raw data and the AI's final recommendation. The *Approve inclusion* and *Exclude* buttons are at the bottom of this column.
- **Right Pane:** A natural-language explanation of the AI's reasoning (e.g., *vulnerability assessment comes out low...*).

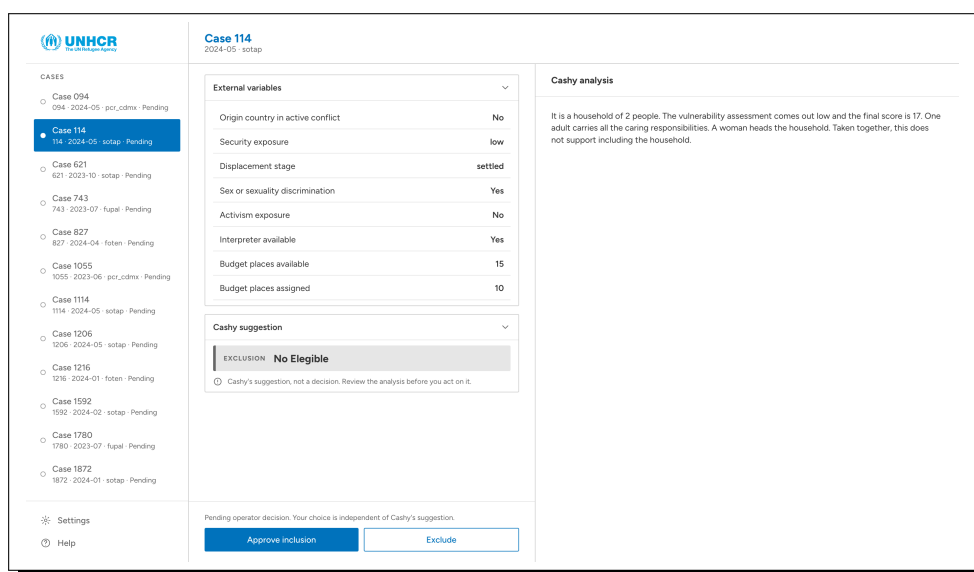


Figure 3: The dashboard's case view.

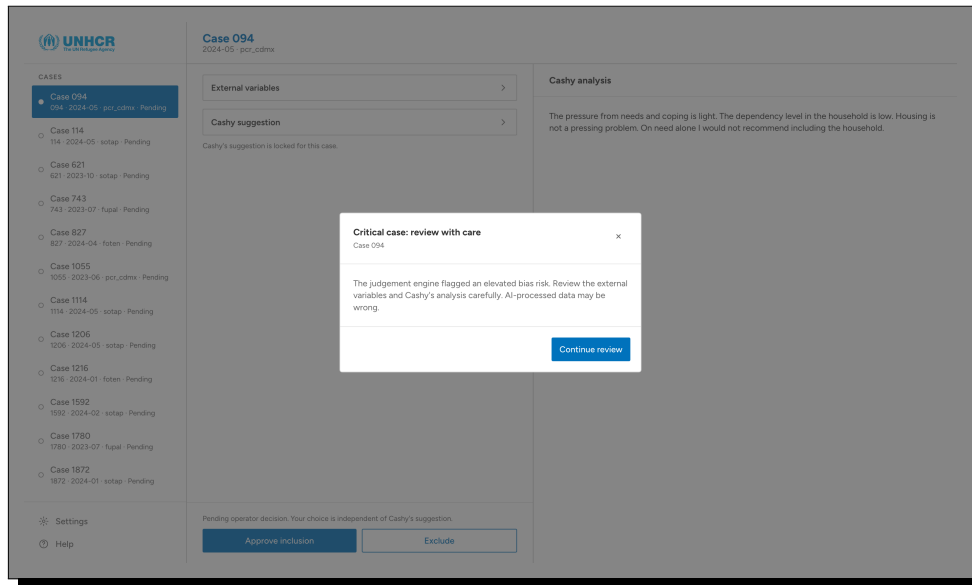


Figure 4: Critical case warning dialog prompting cognitive forcing.

To enforce cognitive engagement and capture decision rationale, the workflow integrates two interactive modal dialogs:

- **Critical Case Warning Modal:** A cognitive forcing mechanism [6] triggered on cases flagged with `show warning: true` when the AI output and scorecard diverge with a low semantic match. By displaying the notice, it forces the operator to slow down and carefully review the data while acknowledging that AI-processed data may be wrong (Figure 4).
- **Operator Survey Modal:** A mandatory feedback form presented upon submitting a decision to sustain active user engagement [5], capturing effectiveness ratings, agreement status, free-text reasoning, and explanation relevance (Figure 5).

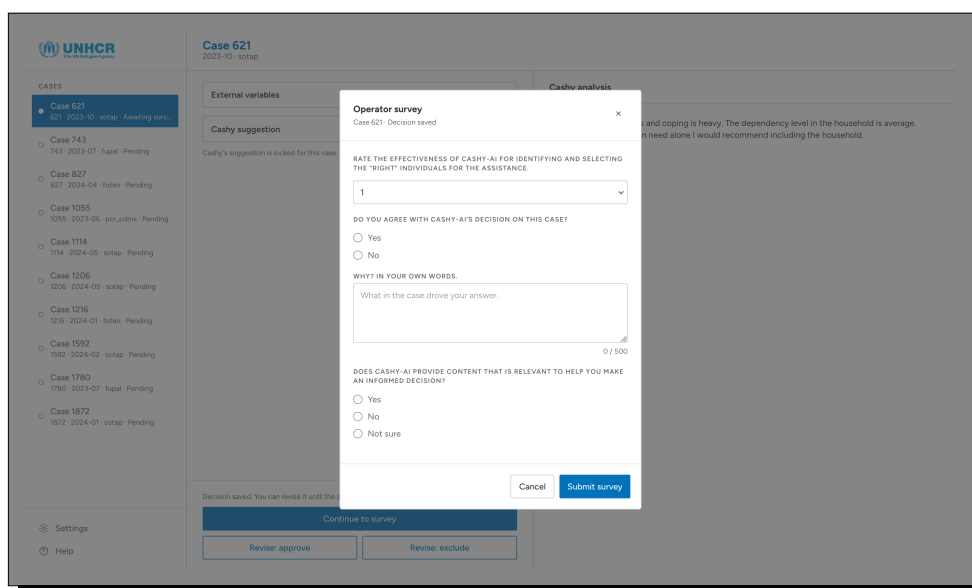


Figure 5: Mandatory operator survey modal capturing decision rationale.

3.3.3 Operator Decision Flow

The oversight workflow structures case processing into sequential operational steps to ensure deliberate human evaluation [5]:

1. **Case Selection:** The operator selects an unreviewed record from the left sidebar queue to initiate evaluation.
2. **Contextual Data Review:** The operator expands the *External variables* accordion to examine household metrics, including country of origin, security exposure, and vulnerability indicators.
3. **AI Recommendation Inspection:** The operator reads Cashy’s natural-language rationale and suggested eligibility status in the right pane.
4. **Independent Judgment:** The operator selects either *Approve inclusion* or *Exclude*, forming a determination independent of automated decision locks.
5. **Mandatory Survey Trigger:** Saving the initial choice launches the structured operator survey modal (Figure 5).
6. **Decision Revision:** The operator maintains the ability to modify or reverse the decision until the survey is formally submitted, ensuring full agency over the final record.

3.3.4 Structured Feedback Loop

To prevent passive acceptance and gather continuous evaluation data, saving a decision automatically triggers a mandatory feedback modal [6]. The survey collects structured inputs on four dimensions:

- **Perceived Effectiveness:** A 1-to-5 scale rating Cashy’s utility in identifying appropriate candidates for assistance.
- **Explicit Agreement:** A binary indicator (*Yes / No*) recording whether the operator concurs with Cashy’s recommendation.
- **Qualitative Rationale:** A mandatory free-text input (500-character limit) documenting the core factors driving the caseworker’s judgment.
- **Explanatory Relevance:** A tri-state evaluation (*Yes / No / Not sure*) assessing whether Cashy’s natural-language explanation provided meaningful context for the case.

4 Conclusion

The Oversight Layer demonstrates that maintaining appropriate human reliance on AI doesn’t require retraining; by embedding cognitive forcing mechanisms, our solution actively mitigates automation bias and algorithm aversion. This lightweight architecture preserves caseworker agency in sensitive cash-targeting decisions while giving institutions the transparency needed to monitor human-AI reliance over time.

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